

ASSIGNMENT 4

Task (1): Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Ans.

RAM Reseating Steps:

1. Turn off the computer and unplug the power cable.
2. Open the computer case and find the RAM slot on the motherboard.
3. Press the clips on both sides of the RAM module.
4. Remove the RAM carefully.
5. Check the RAM and slot for dust or damage.
6. Insert the RAM again into the same slot.

7. Press firmly until the clips lock in place.
8. Close the case and turn on the computer.
9. Check if the computer boots normally.

Task (2):

GPU Troubleshooting Checklist:

- **Check the symptom** – Look for no display, screen flickering, crashes, or low game performance.
- **Check the GPU connection** – Make sure the GPU is properly seated in the PCIe slot.
- **Check power cables** – Ensure the GPU power connectors are firmly connected.
- **Check the display cable** – Make sure the HDMI/DisplayPort cable is connected to the GPU, not the motherboard.

- **Check GPU drivers** – Update or reinstall the graphics driver.
- **Check for overheating** – Make sure the GPU fans are working and clean any dust.
- **Test the PC** – Start a game or graphics test and check for crashes or display problems.
- **Confirm the fix** – If the display works normally and games run without errors, the GPU problem is resolved.

Task (3):

Storage Issue Simulation – Steps:

1. Turn off the PC and unplug the power.
2. Open the case and disconnect the SATA cable from the SSD/HDD.
3. Turn on the PC. The BIOS may show “No Boot Device Found” or the drive may be missing from the boot list.
4. Turn off the PC again and reconnect the

SATA cable firmly.

5. Start the PC and open BIOS/UEFI. The storage drive should appear again.
6. Boot into Windows and check File Explorer/Disk Management to confirm the drive is detected.

- **Before reconnecting:** The BIOS did not detect the storage drive, and the PC showed a boot device error.
- **After reconnecting:** The BIOS detected the drive, and the PC booted normally.

Task (4):

Random PC Restart:

Check first: RAM

Diagnostic Steps

1. Turn off the PC and unplug it.
2. Open the case and check the RAM

for dust or damage.

3. Remove and reinsert the RAM firmly.
 4. Try starting the PC with one RAM stick at a time.
 5. Run Windows Memory Diagnostic to check for RAM errors.
 6. If errors are found, replace the faulty RAM.
 7. Restart the PC and test it by using games or other heavy programs.
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